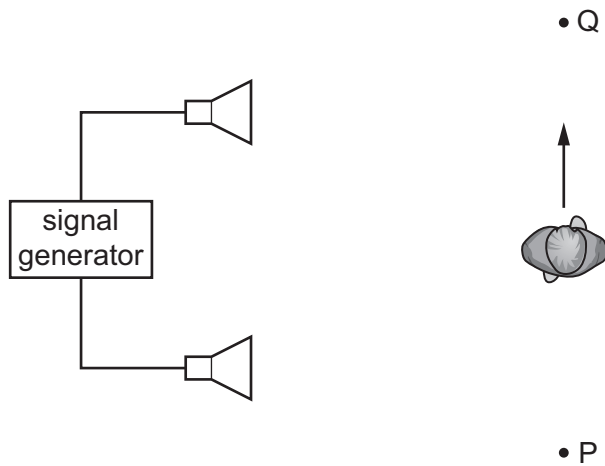


- 31 A student connects two loudspeakers to a signal generator.



As the student walks from P to Q, he notices that the loudness of the sound rises and falls repeatedly.

What causes the loudness of the sound to vary?

- A** diffraction of the sound waves
  - B** Doppler shift of the sound waves
  - C** interference of the sound waves
  - D** reflection of the sound waves
- 32 The current in a resistor of constant resistance is 8.0 mA. The power dissipated by the resistor is  $P$ .

The current in the resistor is increased to 12.0 mA.

What is the new power dissipated by the resistor?

- A**  $0.44P$
- B**  $0.67P$
- C**  $1.5P$
- D**  $2.3P$